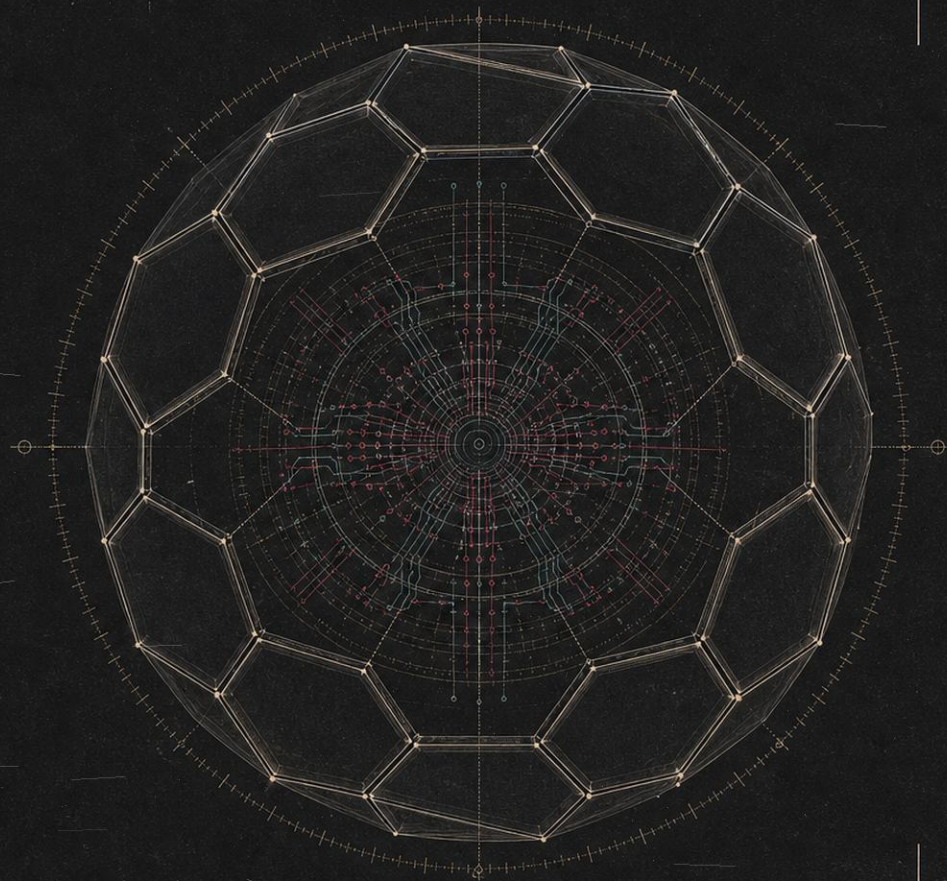


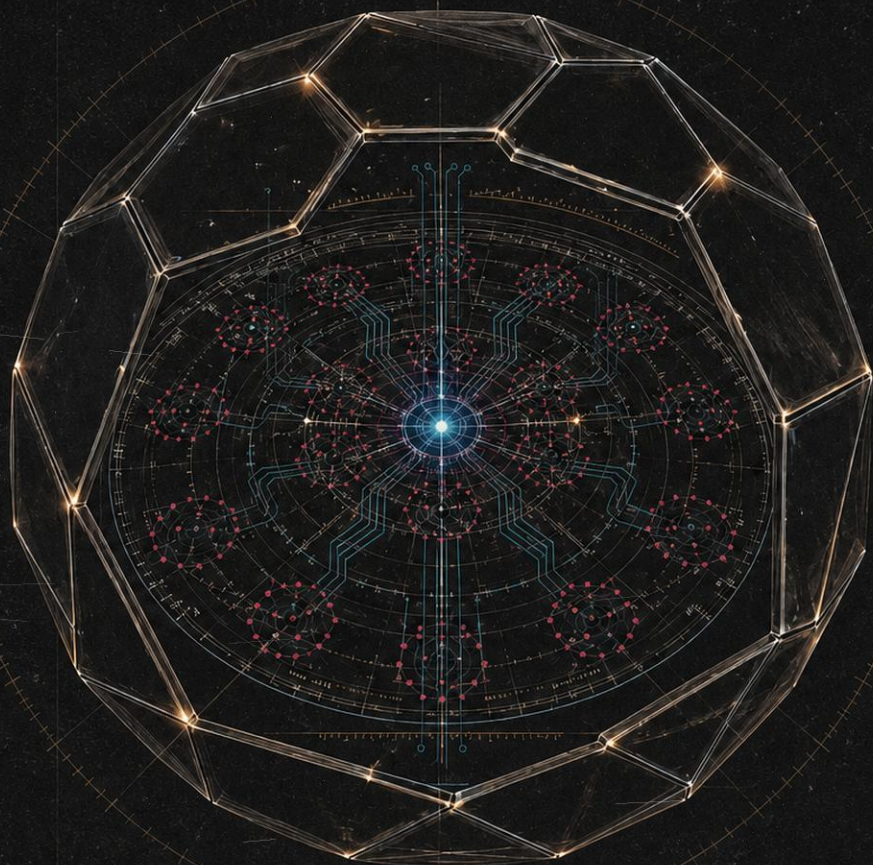
INTRODUCING

AEGIS

VOLUME II

Memory Utility Labs
Calgary, Alberta





a Memory Utility Publication

Introducing

AEGIS

VOLUME II

Memory Utility Labs

Calgary, Alberta

SYSTEM ARCHITECTURE, VOL. II

Product 1.2.0 · Issue 1 · 2026

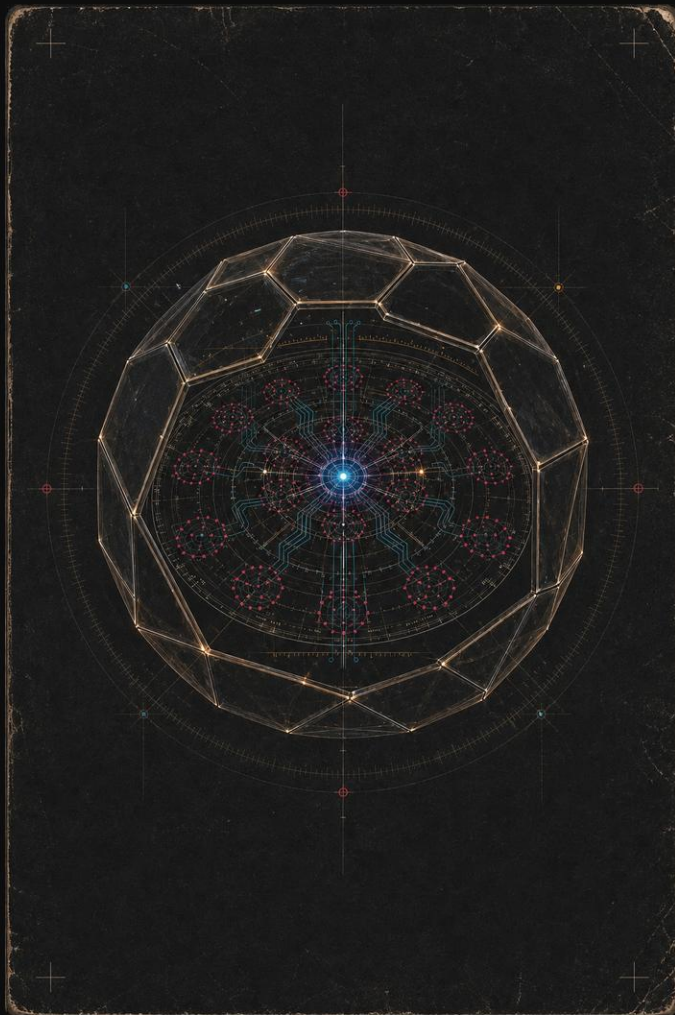
Volume I stated the law. This volume states the machinery.

Control plane. Token capacity basin. Guard middleware. Hermes routing.

Honest yield: savings_percent is billed USD on matched_provider_pairs (D-040). Tiny-chat routing is on. Implement-pack routing is off.

Absolute Form: austere · aggressive on drift · protective reserve ($\geq 80\%$).

GOLDEN GATE BOOK



CONTENTS

This volume

i	Half title	5	III. Guard middleware
ii	Title	6	allow deny require-review
iii	Contents	7	IV. Hermes routing
1	I. Control plane	8	tiny-chat on; implement-pack routing off
2	kernel / process / frozen /v1	9	Colophon
3	II. Token capacity basin		
4	pack / scrub / reserve as low-flow		

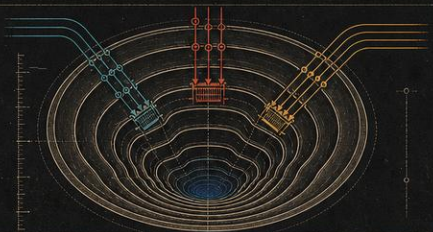
THIS VOLUME

Memory Utility Labs
Calgary, Alberta



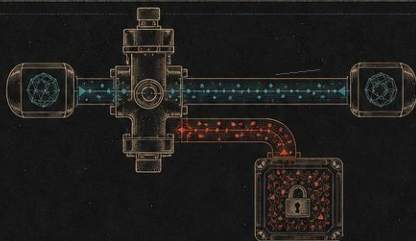
1 CONTROL PLANE

2 TOKEN CAPACITY BASIN

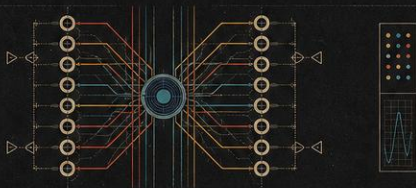


3 GUARD MIDDLEWARE

→ THROUGH-PATH
→ DIVERT PATH



4 HERMES ROUTING SYSTEM



CHAPTER I

Control plane

Kernel (process / memory / drivers / syscalls). Portable data plane under ~/.aegis/. Frozen /v1 API. Honest yield proof.

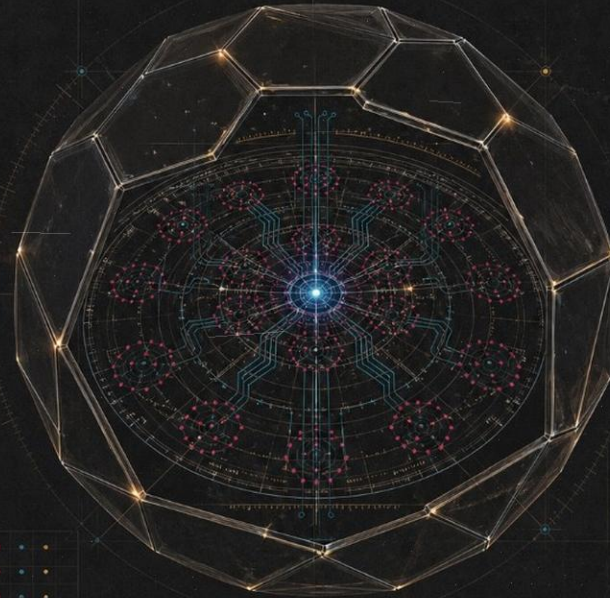
aegis os init lays the plane. aegis doctor --product scores it. aegis os ready is the floor. Optional: aegis serve at <http://127.0.0.1:8787> for OpenAI-compatible calls with body.aegis.pipeline=true.

The serve daemon is not a host kernel. Agent-kernel scores are not Darwin scores. The plane is local. Hosted SaaS, consumer marketplace, and host-kernel replacement are out of scope.

Volume I named the operating system. This chapter is the control surface: process table, frozen contract, local daemon. Nothing here authorizes a remote brain.

AEGIS CONTROL PLANE

Memory Utility Labs
Calgary, Alberta



CHAPTER II

Token capacity basin

Tokens are water. They are not free. Pack and scrub before discharge. A reserve is a low-flow state, not an empty one.

Weekly cap: 1 000 000 processed tokens. Reserve floor: 80%. OPEN / THROTTLE / HARD STOP. Surplus is not decoration: 20% of new savings to the wish jar after audit.

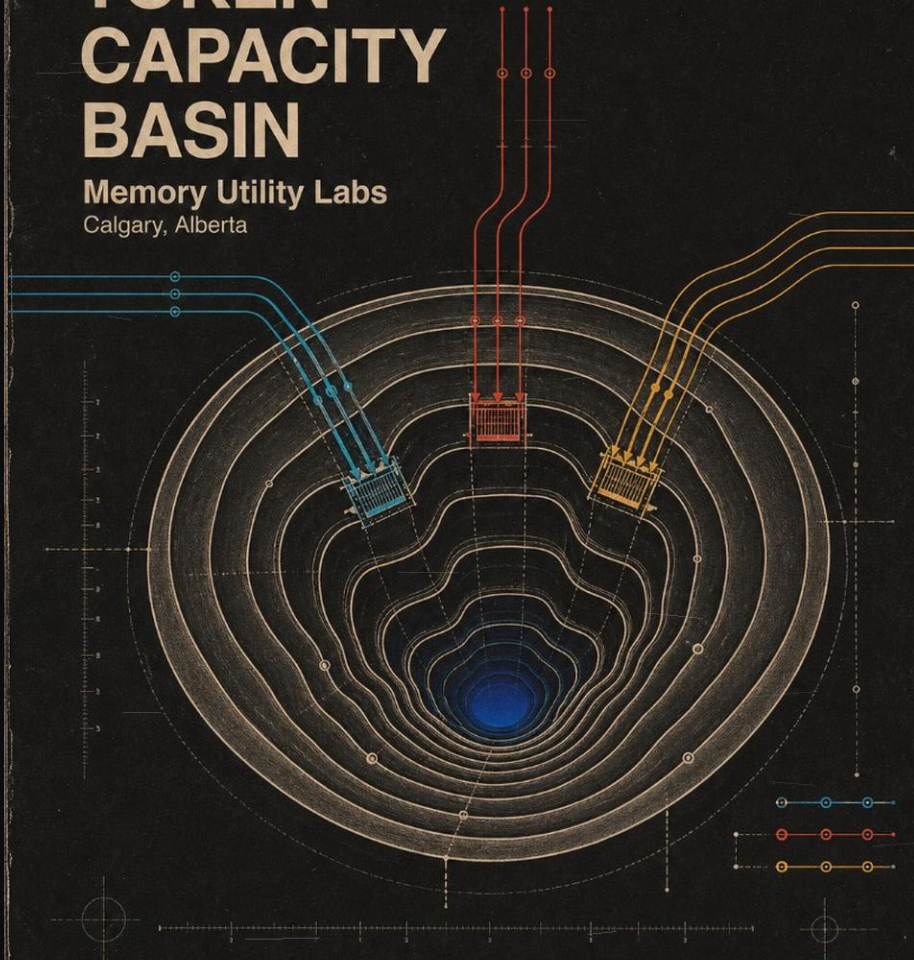
Naïve: dump -> overflow -> re-read. Aegis: pack + reserve + implement-full + land + audit.

The work-system is one basin. Chat windows are jurisdictions, not units. Covering reuse hits if hashes still match. After you edit, the cache misses on purpose.

TOKEN CAPACITY BASIN

Memory Utility Labs
Calgary, Alberta

Publication 20



CHAPTER III

Guard middleware

Single path: request -> normalize -> ids -> classify -> policy -> fail-closed -> allow | deny | require-review -> execute only if permitted -> GuardDecision audit.

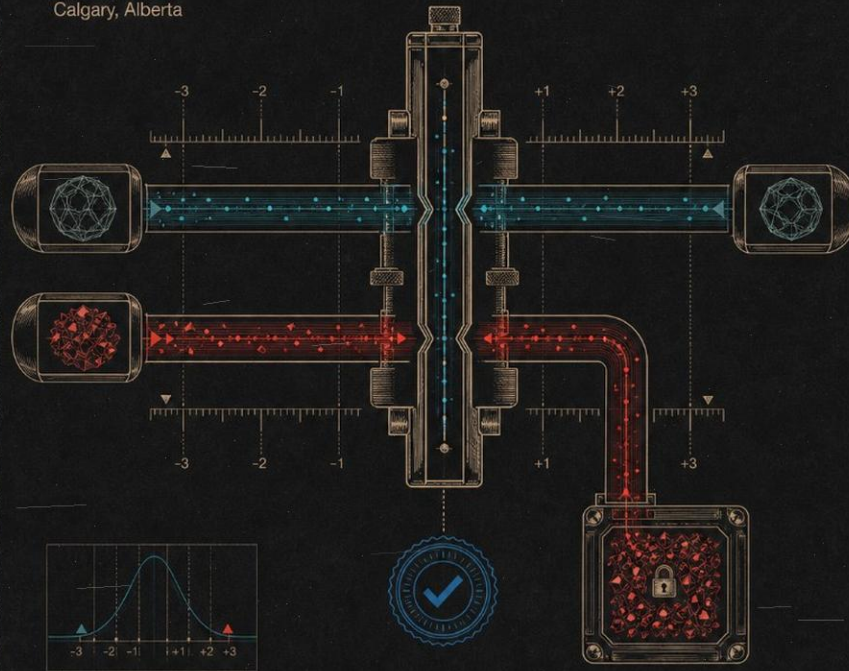
Unknown tools deny. Malformed requests deny. Out-of-scope requests deny. Shadow mode defaults on: the decision is logged; the block is not yet the law. Turn shadow off when the catalog is trusted.

Ledger: `~/aegis/guard_log.jsonl`. Rotate: `aegis guard rotate`. Hermes wrapper never raises to block a tool; it returns a structured deny payload instead.

Hermes middleware itself is fail-open on raise. Guard must not be. Admission is a valve, not a suggestion.

AEGIS GUARD MIDDLEWARE

Memory Utility Labs
Calgary, Alberta



CHAPTER IV

Hermes routing

aegis hermes search|resolve is a thin read-only CLI over the local index. Track first. Perplexity only for a live external miss. Perplexity never edits.

Coordination pathways are drawn. Tiny-chat routing is on (D-040): explore/review v4-pro may swap to nano. Implement packs stay on the requested model. savings_percent is observed billed USD, not a pack guess.

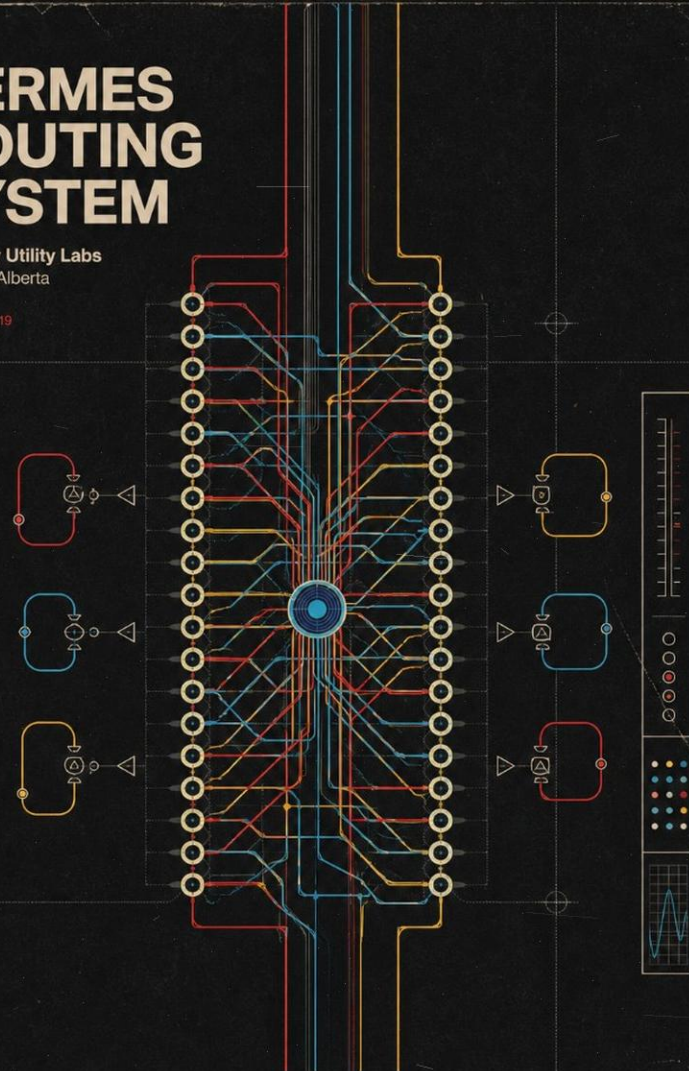
Advertised tool-call metadata is not admission. A research note is not a routing trial. The index is a marshal. It is not a switch.

The evidence scope is bounded: matched tiny-chat pairs authorize a trial, not broad product routing. Implement packs remain on the requested model.

HERMES ROUTING SYSTEM

Memory Utility Labs
Calgary, Alberta

Publication 19



COORDINATION PATHWAYS VOL. II

COLOPHON

Memory Utility Labs

Designed as a Golden Gate Book for Memory Utility Labs, Calgary, Alberta.

Volume II plates follow 1970s NASA / GSFC technical-poster language: charcoal ground, cream Helvetica, muted teal / brick / mustard, drafting marks, analog grain. Chapter figures are the four system posters.

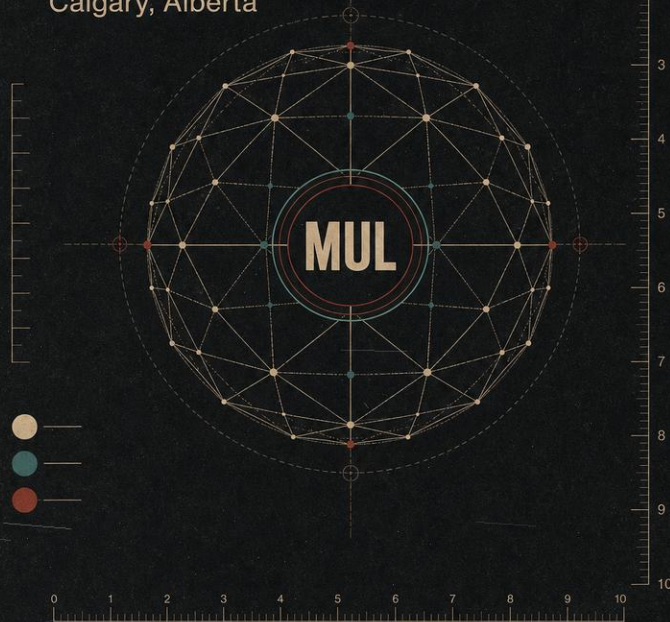
Type: Helvetica. Format: 9 x 6 in landscape, 200 dpi. Product 1.2.0. Issue 1. 2026.

Manuscript from FIELD.md, FIRST_RELEASE.md, D-040. savings_percent is billed USD on matched_provider_pairs.
Tiny-chat routing on. Implement packs unrouted.

Remain in Absolute Form unless explicitly released.

MEMORY UTILITY LABS

Calgary, Alberta



GOLDEN GATE BOOK • VOLUME II • 2026



MEMORY UTILITY LABS
Calgary, Alberta

INTRODUCING AEGIS · VOLUME II · ISSUE 1 · 2026